



COHERENCE

A quantum particle in complete isolation can retain an indeterminate state indefinitely.

ENVIRONMENTAL IMPACT

In a complex system, such as an atom, interference with wave functions from other particles causes indeterminate systems to lose coherence.

ATOM

UNSTABLE ENVIRONMENT

Quantum systems in the indeterminate state described by a wave function are said to be "coherent." The phenomenon of decoherence is a loss of quantum information as a system interacts with its surroundings. Unless a system is perfectly isolated, its coherence deteriorates through interactions with the wave functions of its environment. Decoherence is a major challenge for quantum computing systems that rely on keeping particles in a state of long-term coherence.